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ROTOR STRUCTURE

Abstract

A rotor structure includes a silicon steel main body with a plurality sets of magnetic poles, an axis and an outer edge. Each magnetic pole has a double-layer V-shaped magnet slot structures including an inner V-shaped magnet slot set near the axis and an outer V-shaped magnet slot set near the outer edge. The inner V-shaped magnet slot set has an inner magnetic bridge, and the outer V-shaped magnet slot set has an outer magnetic bridge. The width of the inner magnetic bridge is greater than the width of the outer magnetic bridge. Each magnet slot has two long edges. One of the two long edges that is relatively far away from the corresponding inner magnetic bridge or outer magnetic bridge has a blocking point. The inner V-shaped magnet slot set or the outer V-shaped magnet slot set has a pair of radial magnetic bridges.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Application Ser. No. 63/558,280, filed Feb. 27, 2024 and China Application Serial Number 202422493564.5, filed Oct. 15, 2024, the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND

Field of Disclosure

[0002] The magnets contained in the silicon steel main body of a motor rotor structure have tolerances and lead angles due to manufacturing dimensions. When the magnet slots in the silicon steel main body have no blocking points, there will be a displacement space for the magnets when the rotor is operating.

[0003] In addition, when the position of the magnet blocking point in the silicon steel sheet magnet slot is not proper (e.g., the magnet slot is not provided with a blocking point on the long edge where the inner/outer magnetic bridge is connected), the stress during operation is likely to be concentrated on the silicon steel bridge and causing breakage.

SUMMARY

[0004] The present disclosure provides an improved rotor structure to deal with the needs of the prior art problems.

[0005] In one or more embodiments, a rotor structure including: a silicon steel main body has a plurality sets of magnetic poles, an axis and an outer edge, each set of magnetic pole has a double-layer V-shaped magnet slot structure; wherein the double-layer V-shaped magnet slot structure includes an inner V-shaped magnet slot group near the axis and an outer V-shaped magnet slot group near the outer edge, the inner V-shaped magnet slot group has a first included angle between a pair of inner magnet slots, the outer V-shaped magnet slot group has a second included angle between a pair of outer magnet slots, the first included angle is smaller than the second included angle; wherein the inner V-shaped magnet slot group has an inner magnetic bridge close to the axis along a radial direction, and the outer V-shaped magnet slot group has an outer magnetic bridge close to the axis along the radial direction, a width of the inner magnetic bridge is greater than a width of the outer magnetic bridge; wherein any pair of the pair of inner magnet slots and the pair of outer magnet slots has two long edges, and the one of the two long edges that is relatively far away from the corresponding inner magnetic bridge or the corresponding outer magnetic bridge has a blocking point; wherein the silicon steel main body has a rivet point, and a connection line between a mass center of the rivet point and the axis passes through the inner magnetic bridge and the outer magnetic bridge; and wherein the inner V-shaped magnet slot group or the outer V-shaped magnet slot group has a pair of radial magnetic bridges extending from the pair of inner magnet slots or the pair of outer magnet slots respectively, and the pair of radial magnetic bridges are farther from the axis than the inner magnetic bridge or the outer magnetic bridge.

[0006] In one or more embodiments, a rotor structure including: a silicon steel main body has a plurality sets of magnetic poles, an axis and an outer edge, each set of magnetic pole has a double-layer V-shaped magnet slot structure; wherein the double-layer V-shaped magnet slot structure includes an inner V-shaped magnet slot group near the axis has a pair of inner magnet slots and an

outer V-shaped magnet slot group near the outer edge has a pair of outer magnet slots; wherein the inner V-shaped magnet slot group has an inner magnetic bridge close to the axis along a radial direction, and the outer V-shaped magnet slot group has an outer magnetic bridge close to the axis along the radial direction, a width of the inner magnetic bridge is greater than a width of the outer magnetic bridge; wherein any pair of the pair of inner magnet slots and the pair of outer magnet slots has two long edges, and the one of the two long edges that is relatively far away from the corresponding inner magnetic bridge or the corresponding outer magnetic bridge has a blocking point; and wherein the inner V-shaped magnet slot group or the outer V-shaped magnet slot group has a pair of radial magnetic bridges extending from the pair of inner magnet slots or the pair of outer magnet slots respectively, and the pair of radial magnetic bridges are farther from the axis than the inner magnetic bridge or the outer magnetic bridge.

[0007] In sum, the rotor structure disclosed herein optimizes the designs of the magnetic bridge and adjusts the magnetic bridge in radial extension. The radial magnetic bridge is designed in the same direction as the centrifugal force of the rotor structure's rotating state, which can reduce stress, increase the maximum rotational speed, and effectively reduce the problem of magnetic flux leakage. In addition, a magnetic barrier structure can also be added to the outer edge of the rotor structure to increase the magnetic resistance of the silicon steel sheet path and reduce the performance degradation caused by magnetic flux leakage.

[0008] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

[0010] FIG. 1 illustrates a top view of a single magnetic pole of a motor structure according to a first embodiment of the present disclosure;

[0011] FIG. 2 illustrates a top view of a single magnetic pole of a rotor structure according to a second embodiment of the present disclosure;

[0012] FIG. 3 illustrates a top view of a single magnetic pole of a rotor structure according to a third embodiment of the present disclosure;

[0013] FIG. 4 illustrates a top view of a single magnetic pole of a rotor structure according to a fourth embodiment of the present disclosure;

[0014] FIG. 5 illustrates a top view of a single magnetic pole of a rotor structure according to a fifth embodiment of the present disclosure;

[0015] FIG. 6 illustrates a top view of a single magnetic pole of a rotor structure according to a sixth embodiment of the present disclosure; and

[0016] FIG. 7 illustrates a perspective view of the single magnetic pole of the rotor structure in FIG. 6.

DETAILED DESCRIPTION

[0017] Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

[0018] Reference is made to FIG. 1, which illustrates a top view of a single magnetic pole **100a** of a motor structure according to a first embodiment of the present disclosure. The single magnetic pole **100a** of the rotor structure has an inner V-shaped magnet slot group **103** near an axis **101** and an outer V-shaped magnet slot group **105** near an outer edge **150** to form a double-layer V-shaped

magnet slot structure from inside to outside. There is an air gap width G between the magnetic pole **100a** of the rotor structure and the stator structure **110**. The inner V-shaped magnet slot group **103** includes magnet slot **103a** and magnet slot **103b**, and there is an inner magnetic bridge **102** between the magnet slot **103a** and the magnet slot **103b** along a radial direction RD and close to the axis **101**. The outer V-shaped magnet slot group **105** includes magnet slot **105a** and magnet slot **105b**, and there is an outer magnetic bridge **104** between the magnet slot **105a** and the magnet slot **105b** along the radial direction RD and close to the axis **101**. The width $D1$ of the inner magnetic bridge **102** is greater than the width $D2$ of the outer magnetic bridge **104**. In some embodiments of the present disclosure, the magnetic pole **100a** has a long rectangular magnetic barrier structure (**113a**, **113b**) close to where between the outer V-shaped magnet slot group **105** and the outer edge **150**. A long axis direction of the magnetic barrier structure (**113a**, **113b**) is perpendicular to the outer V-shaped magnet slot group **105**, and can contain air layers, magnets or high magnetic resistance materials. The magnetic pole **100a** has a rivet point **111a** penetrating the silicon steel main body along the axial direction, and a connection line between a geometric center of the rivet point **111a** and the axis **101** passes through the inner magnetic bridge **102** and the outer magnetic bridge **104**.

[0019] In some embodiments of the present disclosure, the width $D2$ of the outer magnetic bridge **104** is equal to the air gap width G between the rotor structure and the stator structure, where the stator structure sleeves the outer edge **150** of the rotor structure along the radial direction RD. In some embodiments of the present disclosure, the width $D1$ of the inner magnetic bridge **102** is 1.5 times the air gap width G between the rotor structure and the stator structure. In some embodiments of the present disclosure, a ratio of the width $D1$ to the width $D2$ ranges from 1.2 to 1.6. In some embodiments of the present disclosure, the ratio of width $D1$ to width $D2$ is equal to 1.5. Rotor structures that comply with the above width relationship have the effect of reducing magnetic bridge stress.

[0020] Reference is made to FIG. 2, which illustrates a top view of a single magnetic pole **100b** of a rotor structure according to a second embodiment of the present disclosure. The single magnetic pole **100b** of the rotor structure also has an inner V-shaped magnet slot group **103** near the axis **101** and an outer V-shaped magnet slot group **105** near the outer edge **150** to form a double-layer V-shaped magnet slot structure. The magnetic pole **100b** also has an inner magnetic bridge **102** and an outer magnetic bridge **104** as well as the above technical features described in previous paragraphs. In some embodiments of the present disclosure, the magnetic pole **100b** has a rivet point **111a** along the axial direction, and a connection between the geometric center of the rivet point **111a** and the axis **101** passes through the inner magnetic bridge **102** and the outer magnetic bridge **104**. The rivet point **111a** is farther away from the axis **101** or closer to the outer edge **150** than the inner magnetic bridge **102** and the outer magnetic bridge **104**. In some embodiments of the present disclosure, the rivet point **111a** is a rectangle, and a long axis direction of the rivet point **111a** is parallel to the radial direction RD. In some embodiments of the present disclosure, the magnetic pole **100b** further has at least one rivet point **111b** between the magnetic poles extending along a normal vector of the magnet surface to reduce the influence on the magnetic force.

[0021] In some embodiments of the present disclosure, the magnet slot **103a**, the magnet slot **103b**, the magnet slot **105a** and the magnet slot **105b** all have blocking points **108**. The blocking points **108** are located on the one of the two long edges of each magnet slot that is relatively far away from the corresponding inner magnetic bridge **102** or the corresponding outer magnetic bridge **104**. In some embodiments of the present disclosure, the magnet **130** accommodated in the magnet slot has a width W (for example, refer to FIG. 6), the blocking point **108** has a height h , and a relationship between the height h of the blocking point **108** and the width W of the magnet **130** is $W/10 \leq h \leq W/5$. This relationship also applies to the height of the blocking point **108** in other embodiments of the present disclosure. In some embodiments of the present disclosure, the magnetic pole **100b** is different from other embodiments in that it has a long rectangular magnetic

barrier structure (**112a**, **112b**) close to and between the outer V-shaped magnet slot group **105** and the outer edge **150**. The long axis direction of the magnetic barrier structure (**112a**, **112b**) is perpendicular to the outer edge **150** or the outer V-shaped magnet slot group **105**, and may contain an air layer, magnet or high magnetic resistance material.

[0022] Reference is made to FIG. 3, which illustrates a top view of a single magnetic pole of a rotor structure according to a third embodiment of the present disclosure. The single magnetic pole **100c** of the rotor structure also has an inner V-shaped magnet slot group **103** near the axis **101** and an outer V-shaped magnet slot group **105** near the outer edge **150** to form a double-layer V-shaped magnet slot structure. The magnetic pole **100c** also has the above-mentioned technical features such as the inner magnetic bridge **102**, the outer magnetic bridge **104** and the blocking point **108** similar to the magnetic pole **100a**. In some embodiments of the present disclosure, the magnetic pole **100c** has a rivet point **111a** penetrating the silicon steel main body along the axial direction, and a connection line between the geometric center of the rivet point **111a** and the axis **101** passes through the inner magnetic bridge **102** and the outer magnetic bridge **104**. In some embodiments of the present disclosure, the magnetic pole **100c** is different from other embodiments in that it has the arc-shaped magnetic barrier structure (**112c**, **112d**), which is configured between the outer V-shaped magnet slot group **105** and the outer edge **150**. A curvature center of the arc-shaped magnetic barrier structure (**112c**, **112d**) generally faces towards the outer V-shaped magnet slot group **105**, and may include an air layer, a magnet, or a high magnetic resistance material.

[0023] Reference is made to FIG. 4, which illustrates a top view of a single magnetic pole **100d** of a rotor structure according to a fourth embodiment of the present disclosure. The single magnetic pole **100d** of the rotor structure also has an inner V-shaped magnet slot group **103** near the axis **101** and an outer V-shaped magnet slot group **105** near the outer edge **150** to form a double-layer V-shaped magnet slot structure. The inner V-shaped magnet slot group **103** includes a magnet slot **103a** and a magnet slot **103b**, and there is an included angle A1 between the magnet slot **103a** and the magnet slot **103b**. The outer V-shaped magnet slot group **105** includes a magnet slot **105a** and a magnet slot **105b**, and there is an included angle A2 between the magnet slot **105a** and the magnet slot **105b**. The included angle A1 is smaller than the included angle A2. The included angle relationship of this double-layer V-shaped magnet slot structure is also applicable to other embodiments of the present disclosure. The magnetic pole **100d** also has the above-mentioned technical features such as the inner magnetic bridge **102**, the outer magnetic bridge **104**, and the blocking point **108**. In some embodiments of the present disclosure, the magnetic pole **100d** has a rivet point **111a** penetrating the silicon steel main body along the axial direction, and a connection line between the geometric center of the rivet point **111a** and the axis **101** passes through the inner magnetic bridge **102** and the outer magnetic bridge **104** (see FIG. 3).

[0024] In some embodiments of the present disclosure, the included angle A1 of the inner V-shaped magnet slot group has the relational expression of $360/p \leq A1 \leq (r-r1) \times 360/p$, where p is quantity of the rotor magnetic pole, r is a radius of the rotor, r1 is a distance between the inner magnetic bridge **102** and the axis **101**. In some embodiments of the present disclosure, the included angle A2 of the outer V-shaped magnet slot group has the relationship formula of $360/p \leq A2 \leq (r-r2) \times 360/p$, where p is quantity of rotor magnetic poles, r is a radius of the rotor, and r2 is a distance between the outer magnetic bridge **104** and the axis **101**. A rotor that conforms to the above relationship of the included angles of the inner and outer magnets can achieve better torque performance. The above-mentioned angle relationship between the inner and outer V-shaped magnet slot groups is also applicable to other embodiments of the present disclosure.

[0025] In some embodiments of the present disclosure, the magnetic pole **100d** is different from other embodiments in that a magnetic barrier structure (**114a**, **114b**) with a semicircular notch is directly formed on the outer edge **150**, which is configured close to the outer V-shaped magnet slot group **105**, and the semicircular notches of the magnetic barrier structure (**114a**, **114b**) are formed along the radial direction RD toward the periphery of the outer edge **150**, and may contain air

layers, magnets or high magnetic resistance materials.

[0026] Reference is made to FIG. 5, which illustrates a top view of a single magnetic pole of a rotor structure according to a fifth embodiment of the present disclosure. The single magnetic pole **100e** of the rotor structure also has an inner V-shaped magnet slot group **103** near the axis **101** and an outer V-shaped magnet slot group **105** near the outer edge **150** to form a double-layer V-shaped magnet slot structure. The magnetic pole **100e** also has the above-mentioned technical features such as the inner magnetic bridge **102**, the outer magnetic bridge **104**, and the blocking point **108**. The magnetic pole **100e** has a rivet point **111c** penetrating the silicon steel main body along the axial direction, and a connection line between the geometric center of the rivet point **111c** and the axis **101** passes through the inner magnetic bridge **102** and the outer magnetic bridge **104**. In some embodiments of the present disclosure, the magnetic pole **100e** is different from other embodiments in that a circular rivet points **111c** is used instead of the aforementioned rectangular rivet points, and the outer V-shaped magnet slot group **105** has a pair of radial magnetic bridges **120a**. The radial magnetic bridge **120a** is farther from the axis **101** than the inner magnetic bridge **102** or the outer magnetic bridge **104**, and the radial magnetic bridge **120a** is closer to the outer edge **150** than the inner magnetic bridge **102** or the outer magnetic bridge **104**. In some embodiments of the present disclosure, the radial magnetic bridge **120a** has a width **D3**, the width **D3** is greater than the air gap width **G** between the rotor structure and the stator structure (referring to FIG. 1), and the width **D3** is less than a thickness **T** of a single silicon steel sheet (referring to FIG. 7). In some embodiments of the present disclosure, rectangular or circular rivet points can be applied to various magnetic poles of the present disclosure. In some embodiments of the present disclosure, the radial magnetic bridge **120a** enables the rotor structure to have better structural stress intensity and lower magnetic leakage characteristics. In some embodiments of the present disclosure, a no-load back electromotive force of a rotor structure with the radial magnetic bridge **120a** (such as the embodiment in FIG. 5) is increased by about 5% compared with a rotor structure without a radial magnetic bridge (such as the embodiment in FIG. 4).

[0027] Reference is made to FIGS. 6 and 7, FIG. 6 illustrates a top view of a single magnetic pole **100f** of a rotor structure according to a sixth embodiment of the present disclosure, and FIG. 7 illustrates a perspective view of the single magnetic pole **100f** of the rotor structure in FIG. 6. A silicon steel main body **119** of the single magnetic pole **100f** is composed of a plurality of silicon steel sheets **119a** stacked in the axial direction. The magnetic poles (e.g., **100a**, **100b**, **100c**, **100d**, **100e**) in the other embodiments are also composed of a plurality of stacked silicon steel sheets. The single magnetic pole **100f** of the rotor structure also has an inner V-shaped magnet slot group **103** near the axis **101** and an outer V-shaped magnet slot group **105** near the outer edge **150** to form a double-layer V-shaped magnet slot structure. The magnetic pole **100f** also has the above-mentioned technical features such as the inner magnetic bridge **102**, the outer magnetic bridge **104**, and the blocking point **108**. In some embodiments of the present disclosure, the magnetic pole **100f** has a rivet point **111a** penetrating the silicon steel main body **119** along the axial direction, and a connection line between the geometric center of the rivet point **111a** and the axis **101** passes through the inner magnetic bridge **102** and the outer magnetic bridge **104**. In some embodiments of the present disclosure, the magnetic pole **100f** is different from other embodiments in that both the inner V-shaped magnet slot group and the outer V-shaped magnet slot group have radial magnetic bridges (**120a**, **120b**). The radial magnetic bridges (**120a**, **120b**) are farther away from the axis **101** than the inner magnetic bridge **102** or the outer magnetic bridge **104**, or the radial magnetic bridges (**120a**, **120b**) are closer to the outer edge **150** than the inner magnetic bridge **102** or the outer magnetic bridge **104**. In some embodiments of the present disclosure, the radial magnetic bridge **120a** has a width **D3** (referring to FIG. 5), the width **D3** is greater than the air gap width **G** between the rotor structure and the stator structure (referring to FIG. 1), and the width **D3** is less than the thickness **T** of a single silicon steel sheet **119a** (referring to FIG. 7). In some embodiments of the present disclosure, the radial magnetic bridge **120b** has a width **D4** (referring to FIG. 6), the width

D4 is greater than the air gap width G between the rotor structure and the stator structure (referring to FIG. 1), and the width D4 is less than the thickness T of a single silicon steel sheet 119a (referring to FIG. 7). In some embodiments of the present disclosure, the radial magnetic bridges (120a, 120b) enable the rotor structure to have better structural stress intensity and lower magnetic leakage characteristics. In some embodiments of the present disclosure, a no-load back electromotive force of the rotor structure with radial magnetic bridges (120a, 120b) (such as the embodiments in FIGS. 6 and 7) is increased by about 7.5% compared with the rotor structure without radial magnetic bridges (such as the embodiment in FIG. 4).

[0028] In sum, the rotor structure disclosed herein optimizes the designs of the magnetic bridge and adjusts the magnetic bridge in radial extension. The radial magnetic bridge is designed in the same direction as the centrifugal force of the rotor structure's rotating state, which can reduce stress, increase the maximum rotational speed, and effectively reduce the problem of magnetic flux leakage. In addition, a magnetic barrier structure can also be added to the outer edge of the rotor structure to increase the magnetic resistance of the silicon steel sheet path and reduce the performance degradation caused by magnetic flux leakage.

[0029] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0030] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A rotor structure comprising: a silicon steel main body has a plurality sets of magnetic poles, an axis and an outer edge, each set of magnetic pole has a double-layer V-shaped magnet slot structure; wherein the double-layer V-shaped magnet slot structure includes an inner V-shaped magnet slot group near the axis and an outer V-shaped magnet slot group near the outer edge, the inner V-shaped magnet slot group has a first included angle between a pair of inner magnet slots, the outer V-shaped magnet slot group has a second included angle between a pair of outer magnet slots, the first included angle is smaller than the second included angle; wherein the inner V-shaped magnet slot group has an inner magnetic bridge close to the axis along a radial direction, and the outer V-shaped magnet slot group has an outer magnetic bridge close to the axis along the radial direction, a width of the inner magnetic bridge is greater than a width of the outer magnetic bridge; wherein any pair of the pair of inner magnet slots and the pair of outer magnet slots has two long edges, and the one of the two long edges that is relatively far away from the corresponding inner magnetic bridge or the corresponding outer magnetic bridge has a blocking point; wherein the silicon steel main body has a rivet point, and a connection line between a geometric center of the rivet point and the axis passes through the inner magnetic bridge and the outer magnetic bridge; and wherein the inner V-shaped magnet slot group or the outer V-shaped magnet slot group has a pair of radial magnetic bridges extending from the pair of inner magnet slots or the pair of outer magnet slots respectively, and the pair of radial magnetic bridges are farther from the axis than the inner magnetic bridge or the outer magnetic bridge.

2. The rotor structure of claim 1, wherein the width of the inner magnetic bridge is 1.5 times an air gap width between the rotor structure and a stator structure.

3. The rotor structure of claim 1, wherein the width of the outer magnetic bridge is equal to an air gap width between the rotor structure and a stator structure.

4. The rotor structure of claim 1, wherein the pair of radial magnetic bridges have a width greater than an air gap width between the rotor structure and a stator structure.

5. The rotor structure of claim 4, wherein the silicon steel main body comprises a plurality of silicon steel sheets stacked along an axial direction, and the width of the pair of radial magnetic bridges is smaller than a thickness of a single one of the silicon steel sheets.
6. The rotor structure of claim 1, further comprising a plurality of magnets disposed in the pair of inner magnet slots and the pair of outer magnet slots, each of the magnets has a width W , and the blocking point has a height h , wherein $W/10 \leq h \leq W/5$.
7. The rotor structure of claim 1, wherein the rivet point has a rectangular shape or a circular shape.
8. The rotor structure of claim 1, wherein the silicon steel main body further includes a plurality of magnetic barrier structures disposed close to where between the outer edge and the pair of outer magnet slots.
9. The rotor structure of claim 8, wherein each magnetic barrier structure is selected from the group consisting of a rectangular magnetic barrier with a long axis direction perpendicular to the outer V-shaped magnet slot group, a rectangular magnetic barrier with a long axis direction perpendicular to the outer edge, an arc-shaped magnetic barrier with a curvature center facing the outer V-shaped magnet slot group, and a semicircular notch magnetic barrier formed along the radial direction toward a periphery of the outer edge.
10. The rotor structure of claim 1, wherein a ratio of the width of the inner magnetic bridge to the width of the outer magnetic bridge ranges from 1.2 to 1.6.
11. A rotor structure comprising: a silicon steel main body has a plurality sets of magnetic poles, an axis and an outer edge, each set of magnetic pole has a double-layer V-shaped magnet slot structure; wherein the double-layer V-shaped magnet slot structure includes an inner V-shaped magnet slot group near the axis has a pair of inner magnet slots and an outer V-shaped magnet slot group near the outer edge has a pair of outer magnet slots; wherein the inner V-shaped magnet slot group has an inner magnetic bridge close to the axis along a radial direction, and the outer V-shaped magnet slot group has an outer magnetic bridge close to the axis along the radial direction, a width of the inner magnetic bridge is greater than a width of the outer magnetic bridge; wherein any pair of the pair of inner magnet slots and the pair of outer magnet slots has two long edges, and the one of the two long edges that is relatively far away from the corresponding inner magnetic bridge or the corresponding outer magnetic bridge has a blocking point; and wherein the inner V-shaped magnet slot group or the outer V-shaped magnet slot group has a pair of radial magnetic bridges extending from the pair of inner magnet slots or the pair of outer magnet slots respectively, and the pair of radial magnetic bridges are farther from the axis than the inner magnetic bridge or the outer magnetic bridge.
12. The rotor structure of claim 11, wherein the width of the inner magnetic bridge is 1.5 times an air gap width between the rotor structure and a stator structure.
13. The rotor structure of claim 11, wherein the width of the outer magnetic bridge is equal to an air gap width between the rotor structure and a stator structure.
14. The rotor structure of claim 11, wherein the pair of radial magnetic bridges have a width greater than an air gap width between the rotor structure and a stator structure.
15. The rotor structure of claim 11, further comprising a plurality of magnets disposed in the pair of inner magnet slots and the pair of outer magnet slots, each of the magnets has a width W , and the blocking point has a height h , wherein $W/10 \leq h \leq W/5$.
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